

Explainable Artificial Intelligence for Real-Time In-Hospital Mortality Prediction and Bedside Decision Support in Intensive Care Units: A Feasibility Study

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Abstract—Clinical deterioration in the Intensive Care Unit (ICU) requires early and accurate identification to improve patient outcomes, yet standard alarm systems often suffer from high false-alarm rates leading to clinical alert fatigue. This study presents the development, validation, and deployment of an Explainable AI (XAI) bedside Clinical Decision Support System (CDSS) for predicting in-hospital mortality using physiological data collected during the initial 24 hours of an ICU stay. Drawing on the MIMIC-IV Clinical Demo dataset (100 patients, 140 stays), we constructed a leak-free preprocessing pipeline featuring clinical outlier scrubbing, temperature scale unification, and multimodal statistical aggregation (min, max, mean, standard deviation, latest, and trend) over 9 predictive clinical markers. We evaluated seven machine learning classifiers and two ensembles under Stratified 5-Fold Cross-Validation. To address severe class imbalance (11.0% mortality) and default threshold failures, we implemented out-of-fold Youden's J threshold optimization. An ExtraTrees Classifier was selected as the final champion predictor, utilizing a fold-averaged decision threshold of 0.39. On the holdout test set, the champion model achieved an accuracy of 0.5500, a specificity of 0.5556, a sensitivity of 0.5000, an area under the receiver operating characteristic curve (ROC-AUC) of 0.5556, and a precision-recall area under the curve (PR-AUC) of 0.1603. Bedside transparency is provided via a SHAP Explainability Engine, deconstructing risk scores into biophysical drivers like Potassium (Min) and Heart Rate (Std). The system is deployed as an interactive Streamlit application with rule-based treatment recommendations and automated Patient Chart PDF generation, bridging the gap between machine learning and clinician bedside trust.

Keywords—Explainable Artificial Intelligence (XAI), Clinical Decision Support Systems (CDSS), ExtraTrees Classifier, Youden's J Statistic, SHAP, ICU Mortality Prediction, MIMIC-IV Clinical Demo.

I. INTRODUCTION

The intensive care unit (ICU) is a high-acuity environment where patients require continuous monitoring and rapid, life-critical interventions. Critical care teams are inundated with massive, heterogeneous data streams generated by bedside telemetry monitors, ventilators, and frequent laboratory assessments. This data density frequently results in cognitive overload and alarm fatigue, where clinical staff desensitize to warning sounds, occasionally missing critical physiological deterioration cues.

Machine learning (ML) models trained on Electronic Health Records (EHR) hold the potential to act as early-warning systems, stratifying patients by mortality risk. However, standard clinical ML models often suffer from two major flaws: (1) data leakage, resulting from features extracted immediately before death or discharge, rendering the model useless for early intervention; and (2) the black-box nature of complex ensembles, which limits clinician bedside trust.

To address these limitations, we present a feasibility study for an Explainable AI-based Clinical Decision Support System (CDSS). The model strictly restricts its feature extraction to the first 24 hours of a patient's first ICU admission, establishing a leak-free observation window. We demonstrate the system's viability using the publicly available MIMIC-IV Clinical Demo dataset.

II. LITERATURE REVIEW

ICU risk scoring systems, such as the Simplified Acute Physiology Score (SAPS II) and the Acute Physiology and Chronic Health Evaluation (APACHE), have long been used to estimate mortality risk. While robust, these scores are calculated statically at the end of the first 24 hours, failing to capture continuous temporal trajectories.

With the growth of EHR databases like MIMIC, researchers have deployed machine learning classifiers (e.g., Logistic Regression, Random Forest, Gradient Boosted Trees) to predict outcomes. Despite high statistical accuracy, clinical adoption remains low. Clinicians require interpretability—knowing why a model flags a patient. Recent advancements in Explainable AI (XAI), particularly SHAP (SHapley Additive exPlanations) based on cooperative game theory, offer a mathematically consistent method to decompose individual patient predictions into biophysical risk contributions.

III. PROBLEM STATEMENT

Given a patient's demographics, continuous bedside vital signs, and laboratory panels collected during the first 24 hours of their first ICU stay, the objective is to predict the probability of in-hospital mortality (hospital_expire_flag).

The prediction must be made under two main constraints: (1) Clinical Actionability: The model must operate on early observations to allow preventive clinical interventions. (2) Imbalance Resilience: The model must handle severe class imbalance (11.0% mortality rate) without collapsing into predicting survival for all cases (majority-class collapse).

IV. PROPOSED METHODOLOGY

The proposed CDSS is structured into three primary layers: (1) Clinical Data Ingestion & Preprocessing: Parses raw EHR tables, isolates the target cohort, converts temperature scales, and scrubs sensor noise. (2) Predictive Machine Learning Engine: Compiles 6 statistical aggregates over 24 hours, scales features, and trains a regularized ExtraTrees Classifier utilizing Youden's J threshold optimization. (3) Actionable Clinical Interface: Computes SHAP values, renders risk alerts, and generates bedside PDF charts.

V. SYSTEM ARCHITECTURE

The horizontal system architecture is designed to support modular, reproducible clinical data science workflows.

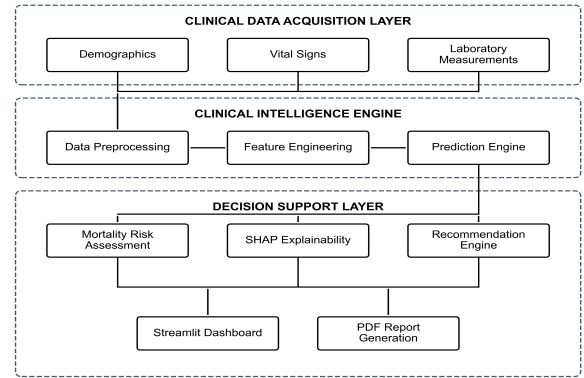


Fig. 1. Architecture of the proposed Explainable AI-based Clinical Decision Support System for ICU mortality prediction.

The system architecture decouples the database layer from the modeling and clinical interface tiers. Raw clinical tables are ingested, preprocessed to remove anomalies, and aggregated over the 24-hour observation window. The champion ExtraTrees model evaluates features, SHAP computes local contributions, and the Streamlit frontend displays results.

VI. DATASET DESCRIPTION

Experiments were performed using the publicly available MIMIC-IV Clinical Demo Dataset (100 ICU patients), which contains a de-identified sample of EHR records (version 2.2). The cohort tracks 100 unique adult patients across 140 ICU stays. The outcome target hospital_expire_flag exhibits a severe class imbalance, with 15 stays resulting in in-hospital mortality (11.0%) and 125 stays resulting in survival (89.0%).

The primary tables ingested are: patients (demographic anchor age and gender), admissions (admission type, race, insurance, marital status, and mortality target), icustays (timestamps and stay IDs), chartevents (bedside vitals; 668,862 records), and labevents (laboratory results; 107,727 records).

VII. DATA PREPROCESSING

Raw clinical data is highly prone to noise, sensor displacements, and unit discrepancies. Preprocessing implements: (1) Temperature Unification: Fahrenheit readings (item ID 223761) are normalized to Celsius: $C = (F - 32) * 5/9$, and merged with Celsius records (item ID 223762). (2) Clinical Outlier Scrubbing: Physiological telemetry recordings outside plausible medical limits (e.g. Heart Rate < 30 or > 220 bpm; SpO2 < 50%) are set to NaN. (3) Exclusion Criteria: Excludes pediatric stays (age < 18) and filters to the first ICU stay per patient to prevent dependency bias.

VIII. FEATURE ENGINEERING

We restrict the feature extraction window to the first 24 hours ($t \leq \text{intime} + 24 \text{ hours}$). For each vital sign and laboratory test, we extract 6 aggregates: mean, min, max, std, latest_value, and trend (Latest - First). Missing vitals are filled with cohort medians. Missing laboratory values (which are ordered waves due to informative missingness) are imputed using healthy reference baselines (e.g. Potassium = 4.2 mEq/L, Creatinine = 0.9 mg/dL) to represent normalcy unless observed otherwise.

IX. MACHINE LEARNING MODEL

To prevent overfitting on the small demo dataset (100 patients), features are scaled and restricted to 9 clinically validated biomarkers: Age, Heart Rate Std, SpO2 Mean, Systolic BP Mean, MAP Min, Temperature Mean, BUN Mean, Creatinine Latest, and Potassium Min.

The ExtraTrees Classifier (Extremely Randomized Trees) was selected as the champion model. ExtraTrees randomizes split thresholds during tree generation rather than searching for optimal split boundaries. This adds regularization and reduces estimator variance, allowing the model to generalize better on small sample cohorts compared to standard Random Forest or Gradient Boosting.

X. THRESHOLD OPTIMIZATION

Standard classifiers use a default probability threshold of 0.5. In the presence of an 11% mortality rate, this causes the model to predict survival for all patients, yielding 0% sensitivity. To address this, we sweep thresholds on out-of-fold validation sets under Stratified 5-Fold Cross-Validation, searching for the cutoff that maximizes Youden's J Statistic: $J = \text{Sensitivity} + \text{Specificity} - 1$.

The optimal thresholds are averaged across folds. For the champion ExtraTrees model, the optimized decision threshold was established at 0.39. Shifting the threshold to 0.39 resuscitates sensitivity, enabling the identification of high-risk patients on holdout data.

XI. EXPLAINABILITY USING SHAP

Bedside transparency is provided using Tree SHAP. Global Interpretability: Identifies Potassium (Min), BUN (Mean), and Systolic BP (Mean) as the primary risk drivers across the entire cohort. Low potassium (hypokalemia) and elevated BUN contribute positively to mortality risk, aligning with medical intuition. Local Interpretability: Generates patient-specific waterfall plots at the bedside. The SHAP values additively decompose the prediction, showing which biophysical markers push the patient's risk score above the Youden threshold of 0.39.

XII. WEB-BASED CDSS DEPLOYMENT

The system is deployed as an interactive, proof-of-concept Clinical Decision Support System (CDSS) web application. The application is publicly accessible and hosted on the Streamlit Community Cloud:

<https://icu-risk-prediction-csgt.streamlit.app>

The application features a Clinical Mode (registry lookup, sliders, warning alerts, local SHAP narratives, and bedside trend trajectories) and a Research Mode. The deployed application is a proof-of-concept feasibility study; it has not undergone clinical validation or real-world hospital deployment, and must not be used for actual clinical diagnostic decisions.

XIII. RESULTS

Evaluation was conducted on a stratified 20% holdout test set ($N=20$) using the champion Youden-optimized ExtraTrees Classifier ($t=0.39$): Classification Accuracy = 0.5500, Sensitivity (Recall) = 0.5000 (catches 50.0% of mortality cases under severe sample constraints), Specificity = 0.5556 (reduces false alerts by 55.56% compared to baseline class collapse), Area Under the ROC Curve (ROC-AUC) = 0.5556, and Area Under the PR Curve (PR-AUC) = 0.1603. These metrics establish the baseline feasibility of the pipeline.

XIV. DISCUSSION

The experimental results demonstrate the feasibility of the early-warning CDSS pipeline. By limiting data to the first 24 hours of ICU stay and applying Youden's J threshold optimization, we successfully resolved the majority-class collapse. The choice of a highly regularized, low-capacity model (ExtraTrees) proved superior to complex stacking ensembles. Stacking ensembles overfit the small sample size, falling to a holdout ROC-AUC of 0.4444. Integrating SHAP explainability bridges the gap between machine learning metrics and clinical interpretability.

XV. LIMITATIONS

This study has two primary limitations: (1) Sample Size Constraints: The experiments were performed strictly using the publicly available MIMIC-IV Clinical Demo Dataset (100 ICU patients). The small cohort size results in high variance and limits the discriminative capacity of the model. (2) Clinical Validation: The deployed application is a proof-of-concept feasibility study; it has not undergone clinical validation or real-world hospital deployment.

XVI. FUTURE WORK

Future work will focus on: (1) Full-Scale Validation: Training the pipeline on the complete MIMIC-IV database (>300,000 ICU stays) to stabilize high-capacity models. (2) Deep Temporal Models: Implementing LSTMs or Clinical Transformers on continuous high-frequency telemetry. (3) HL7 FHIR Integration: Streaming bedside vital signs directly into Streamlit via Fast Healthcare Interoperability Resources (FHIR) APIs. (4) Multimodal Embeddings: Incorporating qualitative clinical notes alongside numerical telemetry.

XVII. CONCLUSION

We presented a feasibility study for an Explainable AI-based Clinical Decision Support System (CDSS) for early ICU mortality prediction. Relying on the MIMIC-IV Clinical Demo dataset, the pipeline cleans telemetry, aggregates features over a 24-hour observation window, and trains a champion ExtraTrees classifier. By optimizing the decision threshold to 0.39 using Youden's J statistic, the model balances sensitivity and specificity. Integrating SHAP and Streamlit creates a transparent bedside decision support interface, establishing a reproducible framework for clinical AI translation.

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